# DATA SCIENCE INTERNSHIP

Week 1 Assignment: E-commerce Sales Analysis

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Generated: 25 October 2026

This report documents the data preparation, validation, key performance indicators, sales trends, category and regional performance, customer insights, and visualizations for the Week 1 Data Science Internship assignment.

Dataset note: The project uses a synthetic educational e-commerce dataset. Its results do not represent actual company performance.

## 1. Dataset Overview

The analysis examines sales records to summarize revenue, orders, products, regions, customers, and payment preferences.

| Metric | Value |
|---|---|
| Raw rows | 662 |
| Raw columns | 11 |
| Cleaned rows | 650 |
| Cleaned columns | 11 |
| Rows used in sales analysis | 623 |

## 2. Data Cleaning and Validation

The raw and cleaned datasets were reviewed for missing values, duplicate rows, inconsistent formats, and invalid values that could affect the analysis.

| Check | Result |
|---|---|
| Exact duplicate rows in raw data | 12 |
| Missing cells before cleaning | 22 |
| Missing cells after cleaning | 7 |
| Rows before cleaning | 662 |
| Rows after cleaning | 650 |

Cleaning decision log: reports/cleaning_decisions.csv

Refer to the notebook's cleaning cells and decision log for the specific rules applied to the dataset.

## 3. Key Performance Indicators

| KPI | Value |
|---|---|
| Total Revenue | 1,088,950 |
| Total Orders | 623 |
| Average Order Value | 1,747.91 |
| Total Units Sold | 1,250 |
| Top-Selling Category (by units) | Electronics |
| Top Revenue-Generating Product | Smart Watch |

| KPI | Value |
|---|---|
| Total analysis revenue | INR 1,088,950.00 |
| Distinct orders | 623 |
| Average order value | INR 1,747.91 |
| Units sold | 1,250 |

## 4. Monthly Sales Trend

Monthly sales analysis helps identify periods of higher and lower revenue. The following table uses the existing monthly analysis output when available.

| Year-Month | Revenue | Units_Sold | Total_Orders | Average_Order_Value |
|---|---|---|---|---|
| 2025-01 | 74,664.00 | 86 | 42 | 1,777.71 |
| 2025-02 | 109,288.00 | 112 | 62 | 1,762.71 |
| 2025-03 | 81,862.00 | 88 | 50 | 1,637.24 |
| 2025-04 | 108,078.00 | 122 | 54 | 2,001.44 |
| 2025-05 | 61,272.00 | 78 | 40 | 1,531.80 |
| 2025-06 | 91,018.00 | 132 | 64 | 1,422.16 |
| 2025-07 | 118,679.00 | 121 | 54 | 2,197.76 |
| 2025-08 | 78,360.00 | 90 | 45 | 1,741.33 |
| 2025-09 | 110,837.00 | 113 | 50 | 2,216.74 |
| 2025-10 | 113,519.00 | 131 | 65 | 1,746.45 |
| 2025-11 | 60,922.00 | 78 | 42 | 1,450.52 |
| 2025-12 | 80,451.00 | 99 | 55 | 1,462.75 |

## 5. Product Category Performance

Category analysis compares revenue and sales volume across product categories. The highest-revenue category may differ from the category with the most units sold.

| Category | Revenue | Units_Sold | Total_Orders | Revenue_Share_Percent |
|---|---|---|---|---|
| Electronics | 430,922.00 | 328 | 167 | 39.57 |
| Fashion | 269,367.00 | 233 | 123 | 24.74 |
| Home & Kitchen | 239,837.00 | 313 | 157 | 22.02 |
| Books | 82,124.00 | 176 | 81 | 7.54 |
| Beauty | 66,700.00 | 200 | 95 | 6.13 |

## 6. Regional Performance

| City | Revenue | Sales_Rows |
|---|---|---|
| Jaipur | 165,868.00 | 87.00 |
| Surat | 165,168.00 | 88.00 |
| Hyderabad | 143,939.00 | 77.00 |
| Pune | 130,959.00 | 73.00 |
| Ahmedabad | 124,063.00 | 72.00 |
| Bengaluru | 123,649.00 | 73.00 |
| Delhi | 115,412.00 | 73.00 |
| Mumbai | 109,403.00 | 74.00 |
| Unknown | 10,489.00 | 6.00 |

The highest-revenue city in the available data is Jaipur, with INR 165,868.00 in revenue.

## 7. Customer Insights

Top identifiable customers by revenue:

| Customer ID | Customer_Revenue | Purchase_Rows |
|---|---|---|
| CUST-1148 | 15,436.00 | 5 |
| CUST-1225 | 13,994.00 | 2 |
| CUST-1014 | 13,790.00 | 3 |
| CUST-1063 | 13,490.00 | 4 |
| CUST-1221 | 13,391.00 | 5 |
| CUST-1240 | 12,891.00 | 3 |
| CUST-1067 | 12,639.00 | 6 |
| CUST-1093 | 12,588.00 | 4 |
| CUST-1217 | 12,239.00 | 4 |
| CUST-1134 | 12,040.00 | 5 |

Customers with more than one sales-analysis row: 170.

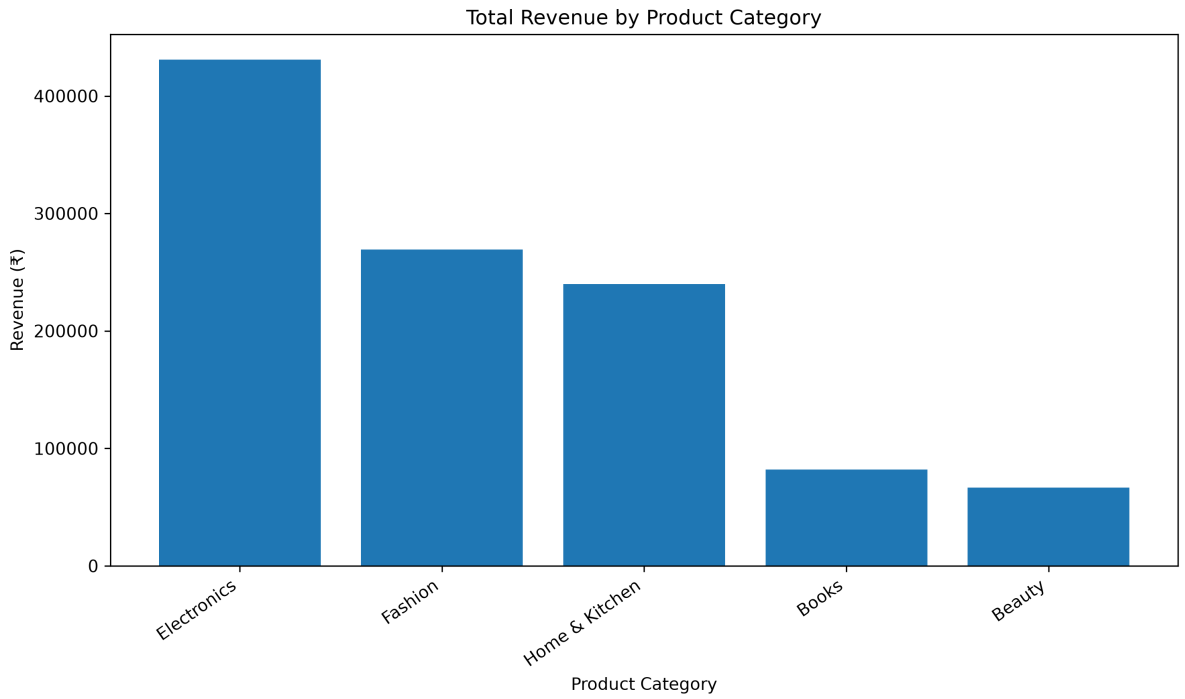
## 8. Customer Revenue Reconciliation

| Revenue component | Amount |
|---|---|
| Total analysis revenue | INR 1,088,950.00 |
| Revenue with identifiable customer IDs | INR 1,068,562.00 |
| Revenue without identifiable customer IDs | INR 20,388.00 |
| Reconciled revenue | INR 1,088,950.00 |

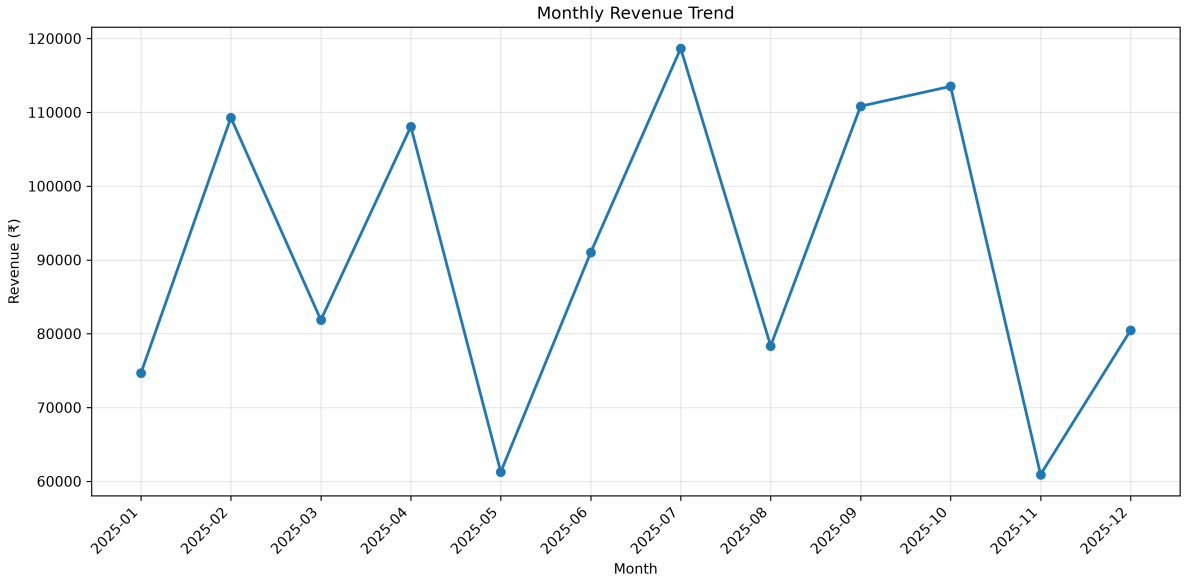
Reconciliation check passed: identified and unidentified customer revenue sum to total revenue.

## 9. Visualizations

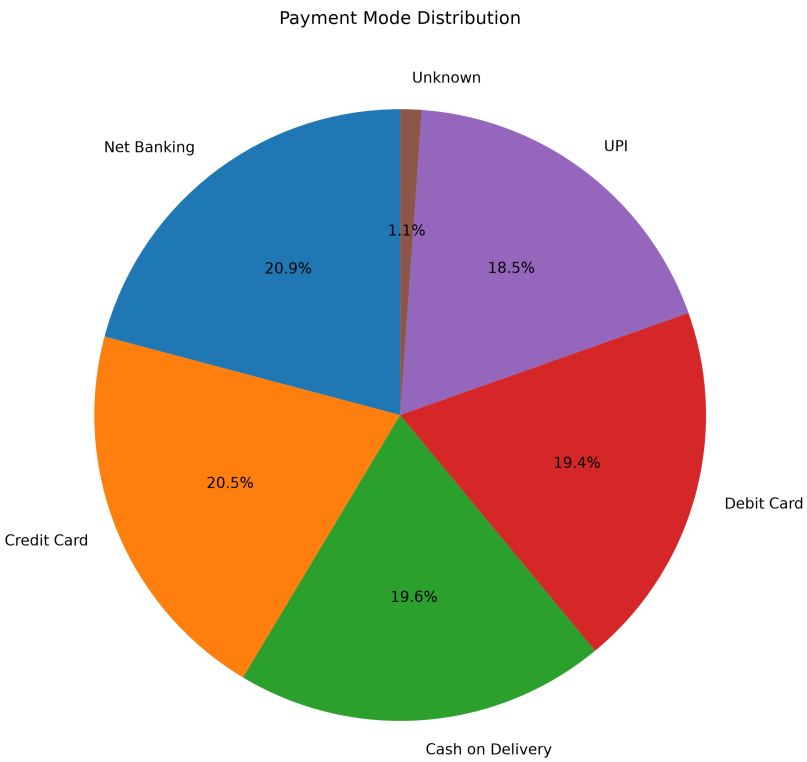
## Revenue by Category



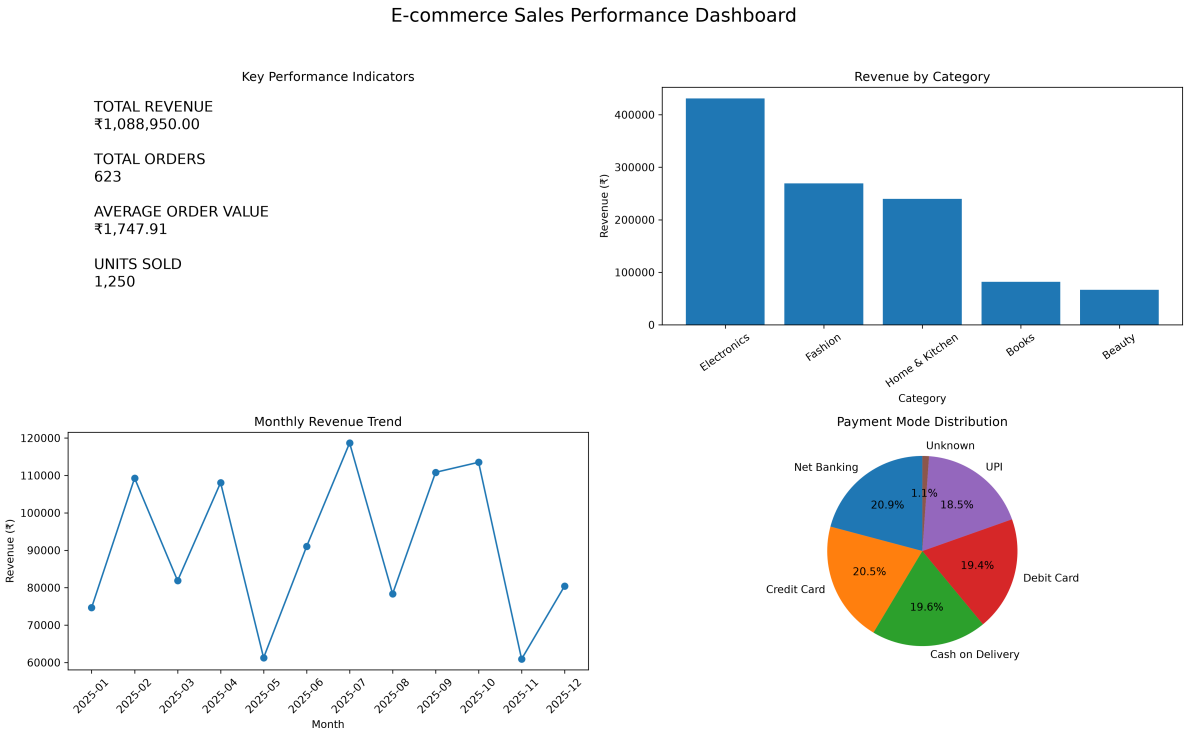

## Monthly Sales Trend



### Payment Mode Share




### Dashboard Overview




## 10. Chart Interpretations

### Revenue by Category

Electronics has the highest recorded category revenue at INR 430,922.00.

Beauty has the lowest recorded category revenue at INR 66,700.00.

### Monthly Sales Trend

Revenue peaked in 2025-07 at INR 118,679.00.

The lowest monthly revenue was in 2025-11 at INR 60,922.00.

### Payment Mode Share

Net Banking is the most frequently recorded payment mode, appearing in 130 rows (20.87% of rows with a payment mode).

## 11. Key Findings

• The sales-analysis dataset contains 623 distinct orders and recorded revenue of INR 1,088,950.00.

• Average order value is INR 1,747.91, calculated as analysis revenue divided by distinct orders.

• Recorded units sold total 1,250.

• Electronics is the highest-revenue category, contributing INR 430,922.00.

• Smart Watch is the highest-revenue product in the analyzed data, with INR 222,411.00.

• The highest-revenue city is Jaipur, with INR 165,868.00.

• The highest-revenue month is 2025-07, with INR 118,679.00.

• Net Banking is the most frequently recorded payment mode, with 130 rows.

## 12. Business Recommendations

• Monitor category revenue and units sold as separate measures.

• Use monthly sales trends to identify periods requiring further investigation.

• Compare regional performance only where reliable location data is available.

• Use customer purchase frequency and revenue together to understand purchasing patterns.

• Validate cleaning rules and KPI definitions before using results for decisions.

## 13. Limitations

• The dataset is synthetic and does not represent actual company performance.

• Missing or unidentified customer IDs limit customer-level analysis.

• Trends and associations do not establish causation.

• Results depend on the cleaning rules and records included in the analysis.

• Regional insights are unavailable if the dataset does not contain a suitable location field.

• Recommendations should be validated against real business data before implementation.

## 14. Reproducibility and Deliverables

To reproduce the analysis, open notebooks/week1_analysis.ipynb and run the cells from top to bottom.

| Deliverable | Status |
|---|---|
| data\raw\ecommerce_sales_raw.csv | Available |
| data\cleaned\ecommerce_sales_cleaned.csv | Available |
| notebooks\week1_analysis.ipynb | Available |
| outputs\kpi_summary.csv | Available |
| outputs\missing_value_report.csv | Available |
| outputs\charts\revenue_by_category.png | Available |

| Deliverable | Status |
|---|---|
| outputs\charts\monthly_sales_trend.png | Available |
| outputs\charts\payment_mode_share.png | Available |
| outputs\dashboard\dashboard_mockup.png | Available |
| README.md | Available |
| requirements.txt | Available |

Review the final report and confirm that its metrics match the notebook outputs before submission.